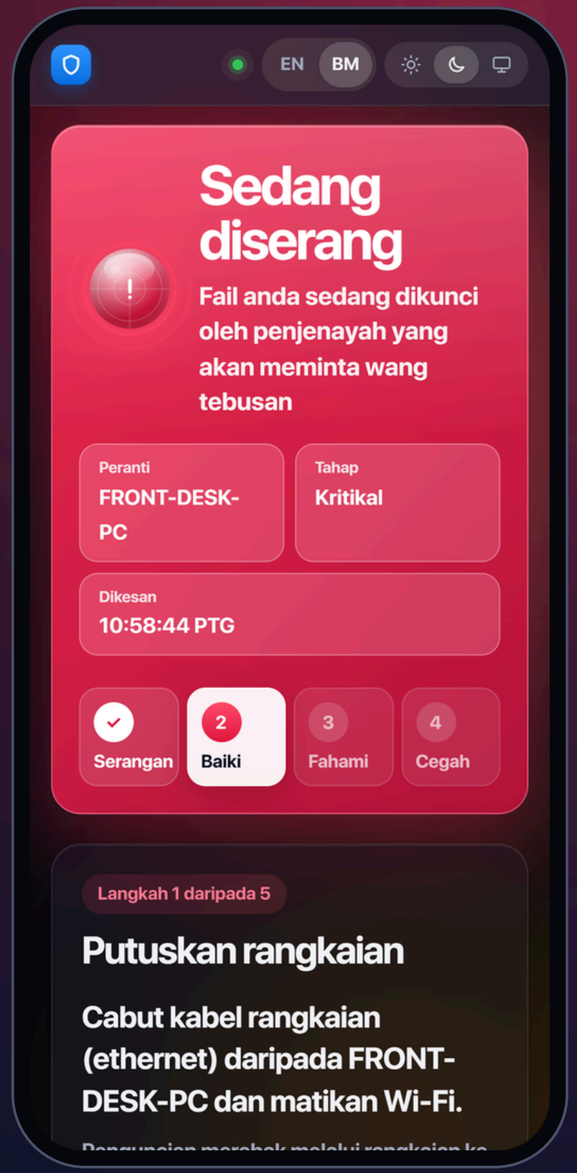
TRACK 4 · CYBERSECURITY & DIGITAL SAFETY

Siaga.

Calm, step-by-step cyber-emergency guidance for Malaysian SMEs.

Panduan kecemasan siber untuk PKS Malaysia

Brand New App



PROJECT & TEAM

Brand New App



Ahmad Zafran

Team leader

Product direction and the pitch



Nabil Ikhwan

Lead developer

Backend, AI guide and deployment



Aerill Danial

Frontend & UX design

Dashboard, user journey and BM interface

Project: Siaga, a cyber-incident guide for small businesses. Built for Track 4: Cybersecurity & Digital Safety.

PROBLEM

When an attack hits, the owner is on their own



Owners can't contain an attack in the first minutes.

12%

of Malaysian SMBs can fix a cyber incident within an hour (Cisco, 2021)

Sources: Bernama (2024), 1,916 elderly victims fall prey to online scams, losing RM255.4 million; Gardaffy, R. (2025), 1,812 kes phishing babitkan kerugian RM65.5 juta, MalaysiaGazette; Cisco (2021), Cybersecurity for SMBs: Asia Pacific Businesses Prepare for Digital Defense.

TARGET USERS & WHY IT MATTERS

Built for the owner, not the IT department

- Malaysian SME owners with no IT staff
- Staff sharing one front-desk PC for invoices, banking and payroll
- Owners more at home in Bahasa Melayu than in technical English
- Older and first-time owners who are new to cyber threats

Why it matters

Small businesses hold invoices, payroll and customer data, but rarely have anyone to call in the first minutes of an attack.

Your first line of defense, even without an IT team. Real-time threat alerts, actionable guidance, and cybersecurity awareness in one personal assistant.

CURRENT LIMITATIONS

Today’s tools warn. They don’t guide.

Antivirus alerts, then stops

It says something is wrong, but not what the owner should physically do next.

Guides are written for IT people

Mostly English, full of jargon, and they assume admin skills the owner doesn’t have.

Help isn’t in the room

Hotlines like Cyber999 help, but someone still has to act on the PC in the first minutes.

The gap isn’t detection. It’s what happens next.

SOLUTION

Four phases. One calm path.



1 Hack

The sensor spots the attack. The screen turns red within a second.



2 Fix

One physical step at a time: unplug it, keep it on, protect the backups.



3 Learn

What happened and how it got in, told as an everyday analogy.



4 Prevent

One-click fixes like 2FA, offline backups and updates. Back to green.

Only the owner’s button presses move the incident forward. The AI explains, it never acts.

UNIQUE VALUE

Why Siaga is different



The AI advises. The owner acts.

The AI can’t press buttons or change anything. Every reply passes safety checks, in English and BM.



Bahasa Melayu, end to end

Every screen, playbook step and AI reply in BM or English, switchable in the middle of an incident.



Works even if the AI doesn’t

Vetted NIST/CISA playbooks answer instantly if the AI is slow or offline. The chat never goes silent.

CORE FEATURES

Everything the owner needs, on one screen

Real-time alerts

Sensor to screen in about 0.3 s in our tests

Guided containment

Steps can’t be skipped or pressed twice

AI emergency guide

Answers questions in plain words, in EN or BM

Learn what happened

How it got in and why it matters to the business

One-click prevention

2FA, offline backups, updates, blocking attackers

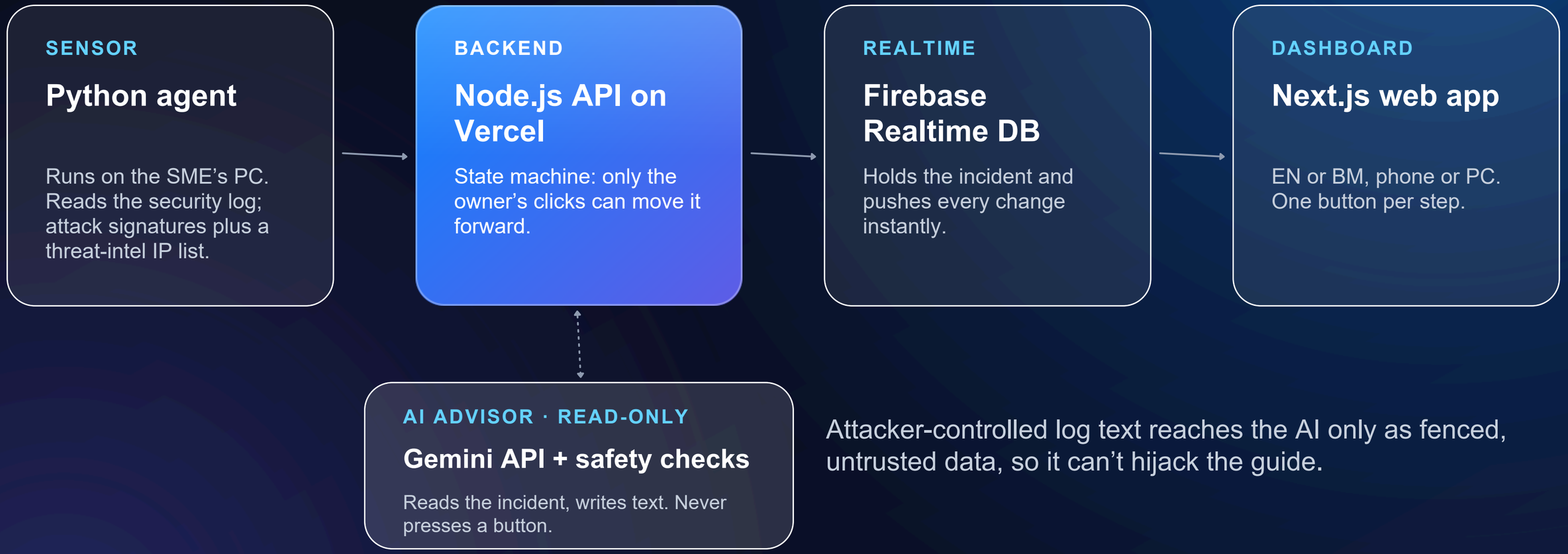
Human escalation

Cyber999 and NSRC 997, one tap away

Recognises ransomware, password guessing, criminal servers and data theft, with a safe universal plan for anything unknown.

TECHNICAL IMPLEMENTATION · ARCHITECTURE

From a log line to the screen in under a second



Attacker-controlled log text reaches the AI only as fenced, untrusted data, so it can't hijack the guide.

TECHNICAL IMPLEMENTATION · STACK

Stack, APIs and integrations

Layer	Technology	Why
Frontend	Next.js (React), English and BM	Works on phone and PC
Backend	Node.js 20, Express, Vercel serverless	Nothing to maintain
Database	Firebase Realtime Database	Live updates to every screen
Sensor	Python 3, standard library only	No extra packages to install
AI	Gemini API (Flash-Lite), read-only, safety checks in EN and BM	Plain-language help that never acts
Playbooks & data	NIST/CISA-based playbooks, malicious-IP list	Vetted steps, threat intelligence
Escalation	Cyber999 (MyCERT), NSRC 997	Human experts one tap away

Tested: 40 automated backend tests and 15 sensor tests.

UX & DESIGN · USER JOURNEY

The owner’s journey, step by step

1

Watching

Green screen. The sensor watches quietly.

2

Alert

The screen turns red within a second and the guide explains.

3

Fix

One physical step, one button. Steps can’t be skipped.

4

Learn

What happened, told as an everyday analogy.

5

All clear

Fixes applied, back to green, time to contain recorded.

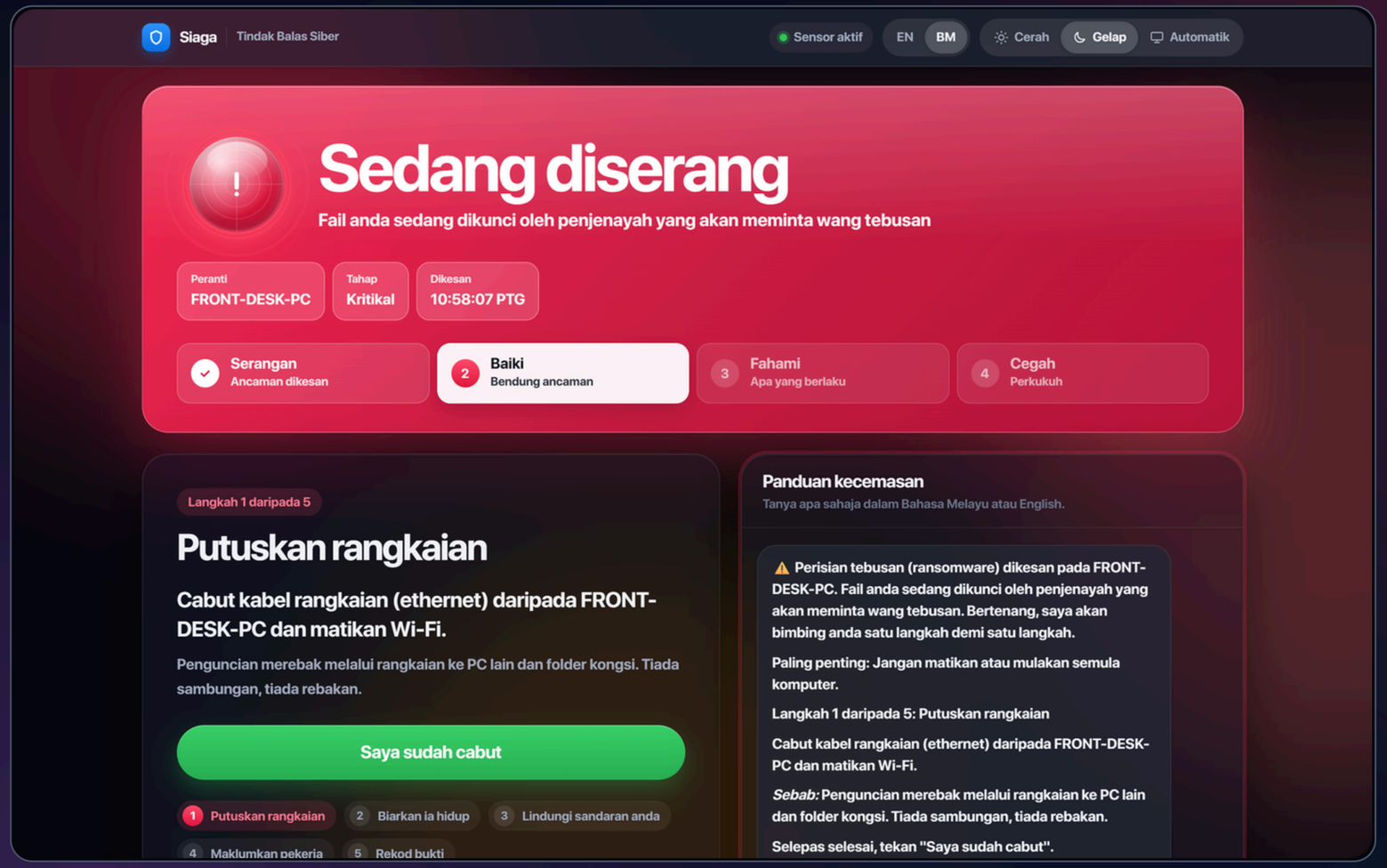
One action per screen

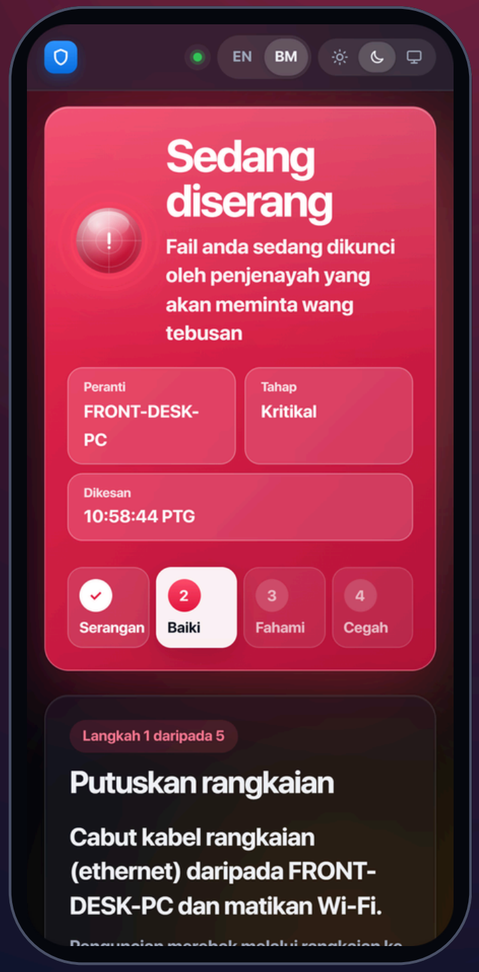
Big buttons, no jargon

EN / BM switch, even mid-incident

UX & DESIGN · PROTOTYPE

Under attack, in Bahasa Melayu



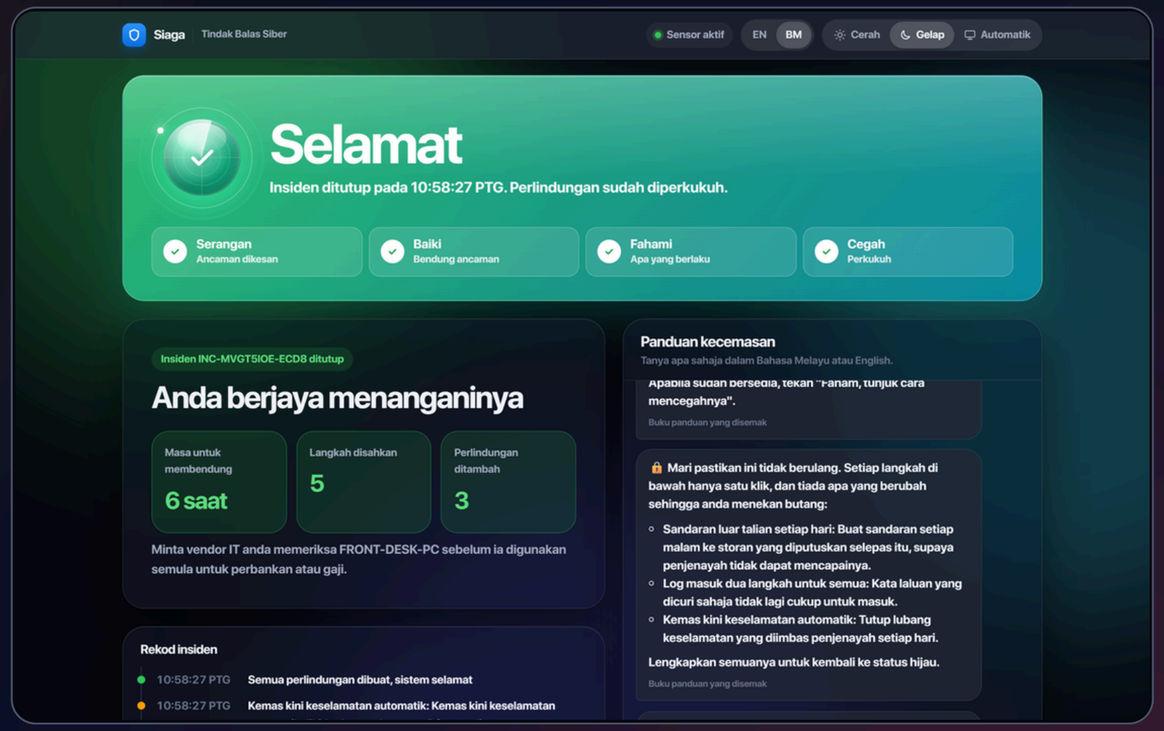


UX & DESIGN · PROTOTYPE

Then it explains, and you’re back to green



Fahami: what happened, in plain words



Selamat: incident closed, time to contain recorded

FEASIBILITY & SCALABILITY

Simple to run today. Ready to grow.

Sustainable today

- Sensor needs only Python, no extra packages
- Serverless backend and Firebase: no servers to manage
- Playbook mode keeps working if the AI is down
- Covered by 55 automated tests

How it scales

- Logins and a private space per business
- Database transactions for many users at once
- Real integrations behind each fix: Google Workspace, Microsoft 365, routers
- A one-click installer for the sensor

IMPACT & ROADMAP

What we measure, and what comes next

Time to contain

Measured automatically for every incident

Steps completed

Proof the owner followed the vetted plan

Protections added

Each incident leaves the business stronger

Pilot goal: 10 SMEs over 3 months. Contain a simulated attack in under 15 minutes, with no IT staff on site.

Next

- WhatsApp alerts and bot (stretch goal)
- MCMC and NSRC scam-database feed (stretch goal)
- Community reporting of attacker addresses (stretch goal)
- Real integrations behind each one-click fix
- Native-speaker review of every BM phrase

THANK YOU · TERIMA KASIH

Siaga.

Calm guidance when it matters most.

Sentiasa siaga.

Code github.com/Wan2Som/Brand-New-App-and-Cybersecurity
Live demo https://siaga-main.vercel.app
Team Brand New App

